

Data Cleaning Report

Titanic Dataset – Data Quality Assessment and Preprocessing

1. Introduction

This project focuses on cleaning and preparing the Titanic dataset for further analysis.

The objective was to identify and address common data quality issues such as missing values, duplicate records, inconsistent formatting, incorrect data types, and potential outliers.

The cleaning process was performed using Python and common data analysis libraries.

2. Dataset

The Titanic dataset was used for this data cleaning task.

The dataset contains passenger information such as Passenger ID, survival status, passenger class, gender, age, number of siblings/spouses aboard, number of parents/children aboard, ticket information, fare, cabin, and port of embarkation.

3. Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

4. Data Quality Assessment

The dataset was first inspected to understand its structure and identify data quality issues.

The following checks were performed:

- Dataset shape
- Column names
- Data types
- Missing values
- Duplicate records
- Descriptive statistics
- Categorical value consistency
- Numerical value ranges

These checks helped identify the main areas that required cleaning.

5. Missing Value Handling

Age

Missing Age values were replaced with the median age. The median was selected because it is less affected by extreme values than the mean.

Embarked

Missing Embarked values were replaced with the most frequently occurring value (mode).

Cabin

Missing Cabin values were replaced with "Unknown" rather than removing the records. This preserves passenger records while indicating that cabin information was unavailable.

6. Duplicate Removal

The dataset was checked for duplicate rows using all available columns. Exact duplicate records were removed where applicable. This helps prevent repeated records from affecting further analysis.

7. Data Standardization

Categorical values were standardized to maintain consistent formatting. The Sex column was standardized so that gender values followed a consistent format. The Embarked column was standardized using uppercase values.

This improves consistency and makes the dataset easier to analyze.

8. Data Type Correction

Data types were reviewed and corrected where necessary. Numerical columns were converted to appropriate numerical data types, while categorical columns such as Sex and Embarked were converted to categorical data types.

This improves data consistency and memory efficiency.

9. Outlier Detection

Potential outliers were identified using the Interquartile Range (IQR) method. The numerical columns examined were Age, SibSp, Parch, and Fare.

Variable	Q1	Q3	IQR	Potential Outliers
Age	22.00	35.00	13.00	66
SibSp	0.00	1.00	1.00	46
Parch	0.00	0.00	0.00	213
Fare	7.91	31.00	23.09	116

Outlier Handling Decision

The identified values were reviewed rather than automatically removed. The potential outliers were retained because many of them can represent genuine passenger characteristics.

Higher Fare values may represent legitimate first-class or premium tickets, while higher SibSp and Parch values may represent passengers travelling with larger families.

The Parch column produced many potential outliers because most passengers had a value of zero. Therefore, the IQR rule was not considered sufficiently informative for automatically removing these observations.

Keeping these values avoids unnecessarily removing valid information from the dataset.

10. Before and After Data Quality

A before-and-after comparison was performed to evaluate the effect of the cleaning process. The comparison included row count, column count, total missing values, duplicate rows, and data type consistency.

Metric	Before Cleaning	After Cleaning
Rows	891	891
Columns	12	12
Total Missing Values	866	0
Duplicate Rows	0	0

The final dataset contains no remaining missing values in the columns addressed during the cleaning process.

11. Visualization

A boxplot was created to visually examine the distribution and potential outliers in the numerical variables.

The visualization is saved in the project images folder as **numerical_outliers.png**.

12. Cleaned Dataset

After completing the cleaning process, the cleaned dataset was saved as **titanic_cleaned.csv**.

The cleaned dataset can be used for further exploratory analysis and machine learning tasks.

13. Conclusion

The Titanic dataset was successfully cleaned and prepared for further analysis.

- Assessing data quality
- Handling missing values
- Removing duplicate records where applicable
- Standardizing categorical values

- Correcting data types
- Detecting potential outliers using the IQR method
- Reviewing and retaining valid extreme observations
- Comparing data quality before and after cleaning
- Saving the cleaned dataset

The final cleaned dataset provides a more consistent and reliable foundation for further analysis of passenger characteristics and survival patterns.